Neural Computation -- 2020

78 article(s), all issues

1. Flexible Working Memory Through Selective Gating and Attentional Tagging

Wouter Kruijne, Sander M. Bohté, Pieter R. Roelfsema, Christian N. L. Olivers

https://doi.org/10.1162/neco_a_01339 | pp. 1-40

Working memory is essential: it serves to guide intelligent behavior of humans and nonhuman primates when task-relevant stimuli are no longer present to the senses. Moreover, complex tasks often require that multiple working memory representations can be flexibly and independently maintained, prioritized, and updated according to changing task demands. Thus far, neural network models of working memory have been unable to offer an integrative account of how such control mechanisms can be acquired in a biologically plausible manner. Here, we present WorkMATe, a neural network architecture that models cognitive control over working memory content and learns the appropriate control operations needed to solve complex working memory tasks. Key components of the model include a gated memory circuit that is controlled by internal actions, encoding sensory information through untrained connections, and a neural circuit that matches sensory inputs to memory content. The network is trained by means of a biologically plausible reinforcement learning rule that relies on attentional feedback and reward prediction errors to guide synaptic updates. We demonstrate that the model successfully acquires policies to solve classical working memory tasks, such as delayed recognition and delayed pro-saccade/anti-saccade tasks. In addition, the model solves much more complex tasks, including the hierarchical 12-AX task or the ABAB ordered recognition task, both of which demand an agent to independently store and updated multiple items separately in memory. Furthermore, the control strategies that the model acquires for these tasks subsequently generalize to new task contexts with novel stimuli, thus bringing symbolic production rule qualities to a neural network architecture. As such, WorkMATe provides a new solution for the neural implementation of flexible memory control.

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2. Conductance-Based Adaptive Exponential Integrate-and-Fire Model

Tomasz Górski, Damien Depannemaecker, Alain Destexhe

https://doi.org/10.1162/neco_a_01342 | pp. 41-66

The intrinsic electrophysiological properties of single neurons can be described by a broad spectrum of models, from realistic Hodgkin-Huxley-type models with numerous detailed mechanisms to the phenomenological models. The adaptive exponential integrate-and-fire (AdEx) model has emerged as a convenient middle-ground model. With a low computational cost but keeping biophysical interpretation of the parameters, it has been extensively used for simulations of large neural networks. However, because of its current-based adaptation, it can generate unrealistic behaviors. We show the limitations of the AdEx model, and to avoid them, we introduce the conductance-based adaptive exponential integrate-and-fire model (CAdEx). We give an analysis of the dynamics of the CAdEx model and show the variety of firing patterns it can produce. We propose the CAdEx model as a richer alternative to perform network simulations with simplified models reproducing neuronal intrinsic properties.

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3. NMDA Receptor Alterations After Mild Traumatic Brain Injury Induce Deficits in Memory Acquisition and Recall

David Gabrieli, Samantha N. Schumm, Nicholas Vigilante, David F. Meaney

https://doi.org/10.1162/neco_a_01343 | pp. 67-95

Mild traumatic brain injury (mTBI) presents a significant health concern with potential persisting deficits that can last decades. Although a growing body of literature improves our understanding of the brain network response and corresponding underlying cellular alterations after injury, the effects of cellular disruptions on local circuitry after mTBI are poorly understood. Our group recently reported how mTBI in neuronal networks affects the functional wiring of neural circuits and how neuronal inactivation influences the synchrony of coupled microcircuits. Here, we utilized a computational neural network model to investigate the circuit-level effects of N-methyl D-aspartate receptor dysfunction. The initial increase in activity in injured neurons spreads to downstream neurons, but this increase was partially reduced by restructuring the network with spike-timing-dependent plasticity. As a model of network-based learning, we also investigated how injury alters pattern acquisition, recall, and maintenance of a conditioned response to stimulus. Although pattern acquisition and maintenance were impaired in injured networks, the greatest deficits arose in recall of previously trained patterns. These results demonstrate how one specific mechanism of cellular-level damage in mTBI affects the overall function of a neural network and point to the importance of reversing cellular-level changes to recover important properties of learning and memory in a microcircuit.

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4. Passive Nonlinear Dendritic Interactions as a Computational Resource in Spiking Neural Networks

Andreas Stöckel, Chris Eliasmith

https://doi.org/10.1162/neco_a_01338 | pp. 96-128

Nonlinear interactions in the dendritic tree play a key role in neural computation. Nevertheless, modeling frameworks aimed at the construction of large-scale, functional spiking neural networks, such as the Neural Engineering Framework, tend to assume a linear superposition of postsynaptic currents. In this letter, we present a series of extensions to the Neural Engineering Framework that facilitate the construction of networks incorporating Dale's principle and nonlinear conductance-based synapses. We apply these extensions to a two-compartment LIF neuron that can be seen as a simple model of passive dendritic computation. We show that it is possible to incorporate neuron models with input-dependent nonlinearities into the Neural Engineering Framework without compromising high-level function and that nonlinear postsynaptic currents can be systematically exploited to compute a wide variety of multivariate, band-limited functions, including the Euclidean norm, controlled shunting, and nonnegative multiplication. By avoiding an additional source of spike noise, the function approximation accuracy of a single layer of two-compartment LIF neurons is on a par with or even surpasses that of two-layer spiking neural networks up to a certain target function bandwidth.

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5. Efficient Actor-Critic Reinforcement Learning With Embodiment of Muscle Tone for Posture Stabilization of the Human Arm

Masami Iwamoto, Daichi Kato

https://doi.org/10.1162/neco_a_01333 | pp. 129-156

This letter proposes a new idea to improve learning efficiency in reinforcement learning (RL) with the actor-critic method used as a muscle controller for posture stabilization of the human arm. Actor-critic RL (ACRL) is used for simulations to realize posture controls in humans or robots using muscle tension control. However, it requires very high computational costs to acquire a better muscle control policy for desirable postures. For efficient ACRL, we focused on embodiment that is supposed to potentially achieve efficient controls in research fields of artificial intelligence or robotics. According to the neurophysiology of motion control obtained from experimental studies using animals or humans, the pedunculopontine tegmental nucleus (PPTn) induces muscle tone suppression, and the midbrain locomotor region (MLR) induces muscle tone promotion. PPTn and MLR modulate the activation levels of mutually antagonizing muscles such as flexors and extensors in a process through which control signals are translated from the substantia nigra reticulata to the brain stem. Therefore, we hypothesized that the PPTn and MLR could control muscle tone, that is, the maximum values of activation levels of mutually antagonizing muscles using different sigmoidal functions for each muscle; then we introduced antagonism function models (AFMs) of PPTn and MLR for individual muscles, incorporating the hypothesis into the process to determine the activation level of each muscle based on the output of the actor in ACRL. ACRL with AFMs representing the embodiment of muscle tone successfully achieved posture stabilization in five joint motions of the right arm of a human adult male under gravity in predetermined target angles at an earlier period of learning than the learning methods without AFMs. The results obtained from this study suggest that the introduction of embodiment of muscle tone can enhance learning efficiency in posture stabilization disorders of humans or humanoid robots.

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6. New Insights Into Learning With Correntropy-Based Regression

Yunlong Feng

https://doi.org/10.1162/neco_a_01334 | pp. 157-173

Stemming from information-theoretic learning, the correntropy criterion and its applications to machine learning tasks have been extensively studied and explored. Its application to regression problems leads to the robustness-enhanced regression paradigm: correntropy-based regression. Having drawn a great variety of successful real-world applications, its theoretical properties have also been investigated recently in a series of studies from a statistical learning viewpoint. The resulting big picture is that correntropy-based regression regresses toward the conditional mode function or the conditional mean function robustly under certain conditions. Continuing this trend and going further, in this study, we report some new insights into this problem. First, we show that under the additive noise regression model, such a regression paradigm can be deduced from minimum distance estimation, implying that the resulting estimator is essentially a minimum distance estimator and thus possesses robustness properties. Second, we show that the regression paradigm in fact provides a unified approach to regression problems in that it approaches the conditional mean, the conditional mode, and the conditional median functions under certain conditions. Third, we present some new results when it is used to learn the conditional mean function by developing its error bounds and exponential convergence rates under conditional ([Formula: see text])-moment assumptions. The saturation effect on the established convergence rates, which was observed under ([Formula: see text])-moment assumptions, still occurs, indicating the inherent bias of the regression estimator. These novel insights deepen our understanding of correntropy-based regression, help cement the theoretic correntropy framework, and enable us to investigate learning schemes induced by general bounded nonconvex loss functions.

7. Associated Learning: Decomposing End-to-End Backpropagation Based on Autoencoders and Target Propagation

Yu-Wei Kao, Hung-Hsuan Chen

https://doi.org/10.1162/neco_a_01335 | pp. 174-193

Backpropagation (BP) is the cornerstone of today's deep learning algorithms, but it is inefficient partially because of backward locking, which means updating the weights of one layer locks the weight updates in the other layers. Consequently, it is challenging to apply parallel computing or a pipeline structure to update the weights in different layers simultaneously. In this letter, we introduce a novel learning structure, associated learning (AL), that modularizes the network into smaller components, each of which has a local objective. Because the objectives are mutually independent, AL can learn the parameters in different layers independently and simultaneously, so it is feasible to apply a pipeline structure to improve the training throughput. Specifically, this pipeline structure improves the complexity of the training time from [Formula: see text], which is the time complexity when using BP and stochastic gradient descent (SGD) for training, to [Formula: see text], where [Formula: see text] is the number of training instances and [Formula: see text] is the number of hidden layers. Surprisingly, even though most of the parameters in AL do not directly interact with the target variable, training deep models by this method yields accuracies comparable to those from models trained using typical BP methods, in which all parameters are used to predict the target variable. Consequently, because of the scalability and the predictive power demonstrated in the experiments, AL deserves further study to determine the better hyperparameter settings, such as activation function selection, learning rate scheduling, and weight initialization, to accumulate experience, as we have done over the years with the typical BP method. In addition, perhaps our design can also inspire new network designs for deep learning. Our implementation is available at https://github.com/SamYWK/Associated_Learning .

8. An EM Algorithm for Capsule Regression

Lawrence K. Saul

https://doi.org/10.1162/neco_a_01336 | pp. 194-226

We investigate a latent variable model for multinomial classification inspired by recent capsule architectures for visual object recognition (Sabour, Frosst, & Hinton, 2017). Capsule architectures use vectors of hidden unit activities to encode the pose of visual objects in an image, and they use the lengths of these vectors to encode the probabilities that objects are present. Probabilities from different capsules can also be propagated through deep multilayer networks to model the part-whole relationships of more complex objects. Notwithstanding the promise of these networks, there still remains much to understand about capsules as primitive computing elements in their own right. In this letter, we study the problem of capsule regression-a higher-dimensional analog of logistic, probit, and softmax regression in which class probabilities are derived from vectors of competing magnitude. To start, we propose a simple capsule architecture for multinomial classification: the architecture has one capsule per class, and each capsule uses a weight matrix to compute the vector of hidden unit activities for patterns it seeks to recognize. Next, we show how to model these hidden unit activities as latent variables, and we use a squashing nonlinearity to convert their magnitudes as vectors into normalized probabilities for multinomial classification. When different capsules compete to recognize the same pattern, the squashing nonlinearity induces nongaussian

terms in the posterior distribution over their latent variables. Nevertheless, we show that exact inference remains tractable and use an expectation-maximization procedure to derive least-squares updates for each capsule's weight matrix. We also present experimental results to demonstrate how these ideas work in practice.

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9. Robust Stability Analysis of Delayed Stochastic Neural Networks via Wirtinger-Based Integral Inequality

R. Suresh, A. Manivannan

https://doi.org/10.1162/neco_a_01344 | pp. 227-243

We discuss stability analysis for uncertain stochastic neural networks (SNNs) with time delay in this letter. By constructing a suitable Lyapunov-Krasovskii functional (LKF) and utilizing Wirtinger inequalities for estimating the integral inequalities, the delay-dependent stochastic stability conditions are derived in terms of linear matrix inequalities (LMIs). We discuss the parameter uncertainties in terms of norm-bounded conditions in the given interval with constant delay. The derived conditions ensure that the global, asymptotic stability of the states for the proposed SNNs. We verify the effectiveness and applicability of the proposed criteria with numerical examples.

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10. Information-Theoretic Representation Learning for Positive-Unlabeled Classification

Tomoya Sakai, Gang Niu, Masashi Sugiyama

https://doi.org/10.1162/neco_a_01337 | pp. 244-268

Recent advances in weakly supervised classification allow us to train a classifier from only positive and unlabeled (PU) data. However, existing PU classification methods typically require an accurate estimate of the class-prior probability, a critical bottleneck particularly for high-dimensional data. This problem has been commonly addressed by applying principal component analysis in advance, but such unsupervised dimension reduction can collapse the underlying class structure. In this letter, we propose a novel representation learning method from PU data based on the information-maximization principle. Our method does not require class-prior estimation and thus can be used as a preprocessing method for PU classification. Through experiments, we demonstrate that our method, combined with deep neural networks, highly improves the accuracy of PU class-prior estimation, leading to state-of-the-art PU classification performance.

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11. Enhanced Signal Detection by Adaptive Decorrelation of Interspike Intervals

William H. Nesse, Leonard Maler, André Longtin

https://doi.org/10.1162/neco_a_01347 | pp. 341-375

Spike trains with negative interspike interval (ISI) correlations, in which long/short ISIs are more likely followed by short/long ISIs, are common in many neurons. They can be described by stochastic models with a spike-triggered adaptation variable. We analyze a phenomenon in these models where such statistically

dependent ISI sequences arise in tandem with quasi-statistically independent and identically distributed (quasi-IID) adaptation variable sequences. The sequences of adaptation states and resulting ISIs are linked by a nonlinear decorrelating transformation. We establish general conditions on a family of stochastic spiking models that guarantee this quasi-IID property and establish bounds on the resulting baseline ISI correlations. Inputs that elicit weak firing rate changes in samples with many spikes are known to be more detectible when negative ISI correlations are present because they reduce spike count variance; this defines a variance-reduced firing rate coding benchmark. We performed a Fisher information analysis on these adapting models exhibiting ISI correlations to show that a spike pattern code based on the quasi-IID property achieves the upper bound of detection performance, surpassing rate codes with the same mean rate-including the variance-reduced rate code benchmark-by 20% to 30%. The information loss in rate codes arises because the benefits of reduced spike count variance cannot compensate for the lower firing rate gain due to adaptation. Since adaptation states have similar dynamics to synaptic responses, the quasi-IID decorrelation transformation of the spike train is plausibly implemented by downstream neurons through matched postsynaptic kinetics. This provides an explanation for observed coding performance in sensory systems that cannot be accounted for by rate coding, for example, at the detection threshold where rate changes can be insignificant.

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12. Deeply Felt Affect: The Emergence of Valence in Deep Active Inference

Casper Hesp, Ryan Smith, Thomas Parr, Micah Allen, Karl Friston, Maxwell J. D. Ramstead

https://doi.org/10.1162/neco_a_01341 | pp. 398-446

The positive-negative axis of emotional valence has long been recognized as fundamental to adaptive behavior, but its origin and underlying function have largely eluded formal theorizing and computational modeling. Using deep active inference, a hierarchical inference scheme that rests on inverting a model of how sensory data are generated, we develop a principled Bayesian model of emotional valence. This formulation asserts that agents infer their valence state based on the expected precision of their action model-an internal estimate of overall model fitness ("subjective fitness"). This index of subjective fitness can be estimated within any environment and exploits the domain generality of second-order beliefs (beliefs about beliefs). We show how maintaining internal valence representations allows the ensuing affective agent to optimize confidence in action selection preemptively. Valence representations can in turn be optimized by leveraging the (Bayes-optimal) updating term for subjective fitness, which we label affective charge (AC). AC tracks changes in fitness estimates and lends a sign to otherwise unsigned divergences between predictions and outcomes. We simulate the resulting affective inference by subjecting an in silico affective agent to a T-maze paradigm requiring context learning, followed by context reversal. This formulation of affective inference offers a principled account of the link between affect, (mental) action, and implicit metacognition. It characterizes how a deep biological system can infer its affective state and reduce uncertainty about such inferences through internal action (i.e., top-down modulation of priors that underwrite confidence). Thus, we demonstrate the potential of active inference to provide a formal and computationally tractable account of affect. Our demonstration of the face validity and potential utility of this formulation represents the first step within a larger research program. Next, this model can be leveraged to test the hypothesized role of valence by fitting the model to behavioral and neuronal responses.

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13. Enhanced Equivalence Projective Simulation: A Framework for Modeling Formation of Stimulus Equivalence Classes

Asieh Abolpour Mofrad, Anis Yazidi, Samaneh Abolpour Mofrad, Hugo L. Hammer, Erik Arntzen

https://doi.org/10.1162/neco_a_01346 | pp. 483-527

Formation of stimulus equivalence classes has been recently modeled through equivalence projective simulation (EPS), a modified version of a projective simulation (PS) learning agent. PS is endowed with an episodic memory that resembles the internal representation in the brain and the concept of cognitive maps. PS flexibility and interpretability enable the EPS model and, consequently the model we explore in this letter, to simulate a broad range of behaviors in matching-to-sample experiments. The episodic memory, the basis for agent decision making, is formed during the training phase. Derived relations in the EPS model that are not trained directly but can be established via the network's connections are computed on demand during the test phase trials by likelihood reasoning. In this letter, we investigate the formation of derived relations in the EPS model using network enhancement (NE), an iterative diffusion process, that yields an offline approach to the agent decision making at the testing phase. The NE process is applied after the training phase to denoise the memory network so that derived relations are formed in the memory network and retrieved during the testing phase. During the NE phase, indirect relations are enhanced, and the structure of episodic memory changes. This approach can also be interpreted as the agent's replay after the training phase, which is in line with recent findings in behavioral and neuroscience studies. In comparison with EPS, our model is able to model the formation of derived relations and other features such as the nodal effect in a more intrinsic manner. Decision making in the test phase is not an ad hoc computational method, but rather a retrieval and update process of the cached relations from the memory network based on the test trial. In order to study the role of parameters on agent performance, the proposed model is simulated and the results discussed through various experimental settings.

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14. Hidden Aspects of the Research ADOS Are Bound to Affect Autism Science

Elizabeth B. Torres, Richa Rai, Sejal Mistry, Brenda Gupta

https://doi.org/10.1162/neco_a_01263 | pp. 515-561

The research-grade Autism Diagnostic Observational Schedule (ADOS) is a broadly used instrument that informs and steers much of the science of autism. Despite its broad use, little is known about the empirical variability inherently present in the scores of the ADOS scale or their appropriateness to define change and its rate, to repeatedly use this test to characterize neurodevelopmental trajectories. Here we examine the empirical distributions of research-grade ADOS scores from 1324 records in a cross-section of the population comprising participants with autism between five and 65 years of age. We find that these empirical distributions violate the theoretical requirements of normality and homogeneous variance, essential for independence between bias and sensitivity. Further, we assess a subset of 52 typical controls versus those with autism and find a lack of proper elements to characterize neurodevelopmental trajectories in a coping nervous system changing at nonuniform, nonlinear rates. Repeating the assessments over four visits in a subset of the participants with autism for whom verbal criteria retained the same appropriate ADOS modules over the time span of the four visits reveals that switching the clinician changes the cutoff scores and consequently influences the diagnosis, despite maintaining fidelity in the same test's modules, room conditions, and tasks' fluidity per visit. Given the changes in probability distribution shape and

dispersion of these ADOS scores, the lack of appropriate metric spaces to define similarity measures to characterize change and the impact that these elements have on sensitivity-bias codependencies and on longitudinal tracking of autism, we invite a discussion on readjusting the use of this test for scientific purposes.

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15. A Novel Neural Model With Lateral Interaction for Learning Tasks

Dequan Jin, Ziyan Qin, Murong Yang, Penghe Chen

https://doi.org/10.1162/neco_a_01345 | pp. 528-551

We propose a novel neural model with lateral interaction for learning tasks. The model consists of two functional fields: an elementary field to extract features and a high-level field to store and recognize patterns. Each field is composed of some neurons with lateral interaction, and the neurons in different fields are connected by the rules of synaptic plasticity. The model is established on the current research of cognition and neuroscience, making it more transparent and biologically explainable. Our proposed model is applied to data classification and clustering. The corresponding algorithms share similar processes without requiring any parameter tuning and optimization processes. Numerical experiments validate that the proposed model is feasible in different learning tasks and superior to some state-of-the-art methods, especially in small sample learning, one-shot learning, and clustering.

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16. Model-Free Robust Optimal Feedback Mechanisms of Biological Motor Control

Tao Bian, Daniel M. Wolpert, Zhong■Ping Jiang

https://doi.org/10.1162/neco_a_01260 | pp. 562-595

Sensorimotor tasks that humans perform are often affected by different sources of uncertainty. Nevertheless, the central nervous system (CNS) can gracefully coordinate our movements. Most learning frameworks rely on the internal model principle, which requires a precise internal representation in the CNS to predict the outcomes of our motor commands. However, learning a perfect internal model in a complex environment over a short period of time is a nontrivial problem. Indeed, achieving proficient motor skills may require years of training for some difficult tasks. Internal models alone may not be adequate to explain the motor adaptation behavior during the early phase of learning. Recent studies investigating the active regulation of motor variability, the presence of suboptimal inference, and model-free learning have challenged some of the traditional viewpoints on the sensorimotor learning mechanism. As a result, it may be necessary to develop a computational framework that can account for these new phenomena. Here, we develop a novel theory of motor learning, based on model-free adaptive optimal control, which can bypass some of the difficulties in existing theories. This new theory is based on our recently developed adaptive dynamic programming (ADP) and robust ADP (RADP) methods and is especially useful for accounting for motor learning behavior when an internal model is inaccurate or unavailable. Our preliminary computational results are in line with experimental observations reported in the literature and can account for some phenomena that are inexplicable using existing models.

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17. Evaluating the Potential Gain of Auditory and Audiovisual Speech-Predictive Coding Using Deep Learning

Thomas Hueber, Eric Tatulli, Laurent Girin, Jean■Luc Schwartz

https://doi.org/10.1162/neco_a_01264 | pp. 596-625

Sensory processing is increasingly conceived in a predictive framework in which neurons would constantly process the error signal resulting from the comparison of expected and observed stimuli. Surprisingly, few data exist on the accuracy of predictions that can be computed in real sensory scenes. Here, we focus on the sensory processing of auditory and audiovisual speech. We propose a set of computational models based on artificial neural networks (mixing deep feedforward and convolutional networks), which are trained to predict future audio observations from present and past audio or audiovisual observations (i.e., including lip movements). Those predictions exploit purely local phonetic regularities with no explicit call to higher linguistic levels. Experiments are conducted on the multispeaker LibriSpeech audio speech database (around 100 hours) and on the NTCD-TIMIT audiovisual speech database (around 7 hours). They appear to be efficient in a short temporal range (25-50 ms), predicting 50% to 75% of the variance of the incoming stimulus, which could result in potentially saving up to three-quarters of the processing power. Then they quickly decrease and almost vanish after 250 ms. Adding information on the lips slightly improves predictions, with a 5% to 10% increase in explained variance. Interestingly the visual gain vanishes more slowly, and the gain is maximum for a delay of 75 ms between image and predicted sound.

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18. Switching in Cerebellar Stellate Cell Excitability in Response to a Pair of Inhibitory/Excitatory Presynaptic Inputs: A Dynamical System Perspective

Saeed Farjami, Ryan P.D. Alexander, Derek Bowie, Anmar Khadra

https://doi.org/10.1162/neco_a_01261 | pp. 626-658

Cerebellar stellate cells form inhibitory synapses with Purkinje cells, the sole output of the cerebellum. Upon stimulation by a pair of varying inhibitory and fixed excitatory presynaptic inputs, these cells do not respond to excitation (i.e., do not generate an action potential) when the magnitude of the inhibition is within a given range, but they do respond outside this range. We previously used a revised Hodgkin–Huxley type of model to study the nonmonotonic first-spike latency of these cells and their temporal increase in excitability in whole cell configuration (termed run-up). Here, we recompute these latency profiles using the same model by adapting an efficient computational technique, the two-point boundary value problem, that is combined with the continuation method. We then extend the study to investigate how switching in responsiveness, upon stimulation with presynaptic inputs, manifests itself in the context of run-up. A three-dimensional reduced model is initially derived from the original six-dimensional model and then analyzed to demonstrate that both models exhibit type 1 excitability possessing a saddle-node on an invariant cycle (SNIC) bifurcation when varying the amplitude of [Formula: see text]. Using slow-fast analysis, we show that the original model possesses three equilibria lying at the intersection of the critical manifold of the fast subsystem and the nullcline of the slow variable [Formula: see text] (the inactivation of the A-type K[Formula: see text] channel), the middle equilibrium is of saddle type with two-dimensional stable manifold (computed from the reduced model) acting as a boundary between the responsive and non-responsive regimes, and the (ghost of) SNIC is formed when the [Formula: see text]-nullcline is (nearly) tangential to the critical manifold. We also show that the slow dynamics associated with (the ghost of) the SNIC and the lower stable branch of the critical manifold are responsible for generating the nonmonotonic first-spike latency. These results thus provide

important insight into the complex dynamics of stellate cells.

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19. Classification from Triplet Comparison Data

Zhenghang Cui, Nontawat Charoenphakdee, Issei Sato, Masashi Sugiyama

https://doi.org/10.1162/neco_a_01262 | pp. 659-681

Learning from triplet comparison data has been extensively studied in the context of metric learning, where we want to learn a distance metric between two instances, and ordinal embedding, where we want to learn an embedding in a Euclidean space of the given instances that preserve the comparison order as much as possible. Unlike fully labeled data, triplet comparison data can be collected in a more accurate and human-friendly way. Although learning from triplet comparison data has been considered in many applications, an important fundamental question of whether we can learn a classifier only from triplet comparison data without all the labels has remained unanswered. In this letter, we give a positive answer to this important question by proposing an unbiased estimator for the classification risk under the empirical risk minimization framework. Since the proposed method is based on the empirical risk minimization framework, it inherently has the advantage that any surrogate loss function and any model, including neural networks, can be easily applied. Furthermore, we theoretically establish an estimation error bound for the proposed empirical risk minimizer. Finally, we provide experimental results to show that our method empirically works well and outperforms various baseline methods.

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20. Center Manifold Analysis of Plateau Phenomena Caused by Degeneration of Three-Layer Perceptron

Daiji Tsutsui

https://doi.org/10.1162/neco_a_01268 | pp. 683-710

A hierarchical neural network usually has many singular regions in the parameter space due to the degeneration of hidden units. Here, we focus on a three-layer perceptron, which has one-dimensional singular regions comprising both attractive and repulsive parts. Such a singular region is often called a Milnor-like attractor. It is empirically known that in the vicinity of a Milnor-like attractor, several parameters converge much faster than the rest and that the dynamics can be reduced to smaller-dimensional ones. Here we give a rigorous proof for this phenomenon based on a center manifold theory. As an application, we analyze the reduced dynamics near the Milnor-like attractor and study the stochastic effects of the online learning.

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21. Neural Model of Coding Stimulus Orientation and Adaptation

Henrikas Vaitkevi■ius, Algimantas Šv■gžda, Rytis Stanik■nas, Remigijus Bliumas, Alvydas Šoli■nas, Janus J. Kulikowski

https://doi.org/10.1162/neco_a_01269 | pp. 711-740

The coding of line orientation in the visual system has been investigated extensively. During the prolonged viewing of a stimulus, the perceived orientation continuously changes (normalization effect). Also, the orientation of the adapting stimulus and the background stimuli influence the perceived orientation of the

subsequently displayed stimulus: tilt after-effect (TAE) or tilt illusion (TI). The neural mechanisms of these effects are not fully understood. The proposed model includes many local analyzers, each consisting of two sets of neurons. The first set has two independent cardinal detectors (CDs), whose responses depend on stimulus orientation. The second set has many orientation detectors (OD) tuned to different orientations of the stimulus. The ODs sum up the responses of the two CDs with respective weightings and output a preferred orientation depending on the ratio of CD responses. It is suggested that during prolonged viewing, the responses of the CDs decrease: the greater the excitation of the detector, the more rapid the decrease in its response. Thereby, the ratio of CD responses changes during the adaptation, causing the normalization effect and the TAE. The CDs of the different local analyzers laterally inhibit each other and cause the TI. We show that the properties of this model are consistent with both psychophysical and neurophysiological findings related to the properties of orientation perception, and we investigate how these mechanisms can affect the orientation's sensitivity.

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22. Feature Extraction of Surface Electromyography Based on Improved Small-World Leaky Echo State Network

Xugang Xi, Wenjun Jiang, Seyed M. Miran, Xian Hua, Yun■Bo Zhao, Yang Chen, Zhizeng Luo

https://doi.org/10.1162/neco_a_01270 | pp. 741-758

Surface electromyography (sEMG) is an electrophysiological reflection of skeletal muscle contractile activity that can directly reflect neuromuscular activity. It has been a matter of research to investigate feature extraction methods of sEMG signals. In this letter, we propose a feature extraction method of sEMG signals based on the improved small-world leaky echo state network (ISWLESN). The reservoir of leaky echo state network (LESN) is connected by a random network. First, we improved the reservoir of the echo state network (ESN) by these networks and used edge-added probability to improve these networks. That idea enhances the adaptability of the reservoir, the generalization ability, and the stability of ESN. Then we obtained the output weight of the network through training and used it as features. We recorded the sEMG signals during different activities: falling, walking, sitting, squatting, going upstairs, and going downstairs. Afterward, we extracted corresponding features by ISWLESN and used principal component analysis for dimension reduction. At the end, scatter plot, the class separability index, and the Davies-Bouldin index were used to assess the performance of features. The results showed that the ISWLESN clustering performance was better than those of LESN and ESN. By support vector machine, it was also revealed that the performance of ISWLESN for classifying the activities was better than those of ESN and LESN.

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23. Online Learning Based on Online DCA and Application to Online Classification

Hoai An Le Thi, Vinh Thanh Ho

https://doi.org/10.1162/neco_a_01266 | pp. 759-793

We investigate an approach based on DC (Difference of Convex functions) programming and DCA (DC Algorithm) for online learning techniques. The prediction problem of an online learner can be formulated as a DC program for which online DCA is applied. We propose the two so-called complete/approximate versions of online DCA scheme and prove their logarithmic/sublinear regrets. Six online DCA-based algorithms are developed for online binary linear classification. Numerical experiments on a variety of benchmark

classification data sets show the efficiency of our proposed algorithms in comparison with the state-of-the-art online classification algorithms.

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24. Optimal Multivariate Tuning with Neuron-Level and Population-Level Energy Constraints

Yuval Harel, Ron Meir

https://doi.org/10.1162/neco_a_01267 | pp. 794-828

Optimality principles have been useful in explaining many aspects of biological systems. In the context of neural encoding in sensory areas, optimality is naturally formulated in a Bayesian setting as neural tuning which minimizes mean decoding error. Many works optimize Fisher information, which approximates the minimum mean square error (MMSE) of the optimal decoder for long encoding time but may be misleading for short encoding times. We study MMSE-optimal neural encoding of a multivariate stimulus by uniform populations of spiking neurons, under firing rate constraints for each neuron as well as for the entire population. We show that the population-level constraint is essential for the formulation of a well-posed problem having finite optimal tuning widths and optimal tuning aligns with the principal components of the prior distribution. Numerical evaluation of the two-dimensional case shows that encoding only the dimension with higher variance is optimal for short encoding times. We also compare direct MMSE optimization to optimization of several proxies to MMSE: Fisher information, maximum likelihood estimation error, and the Bayesian Cramér-Rao bound. We find that optimization of these measures yields qualitatively misleading results regarding MMSE-optimal tuning and its dependence on encoding time and energy constraints.

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25. A Survey on Deep Learning for Multimodal Data Fusion

Jing Gao, Peng Li, Zhikui Chen, Jianing Zhang

https://doi.org/10.1162/neco_a_01273 | pp. 829-864

With the wide deployments of heterogeneous networks, huge amounts of data with characteristics of high volume, high variety, high velocity, and high veracity are generated. These data, referred to multimodal big data, contain abundant intermodality and cross-modality information and pose vast challenges on traditional data fusion methods. In this review, we present some pioneering deep learning models to fuse these multimodal big data. With the increasing exploration of the multimodal big data, there are still some challenges to be addressed. Thus, this review presents a survey on deep learning for multimodal data fusion to provide readers, regardless of their original community, with the fundamentals of multimodal deep learning fusion method and to motivate new multimodal data fusion techniques of deep learning. Specifically, representative architectures that are widely used are summarized as fundamental to the understanding of multimodal deep learning. Then the current pioneering multimodal data fusion deep learning models are summarized. Finally, some challenges and future topics of multimodal data fusion deep learning models are described.

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26. Performance Limitations in Sensorimotor Control: Trade-Offs Between Neural Computation and Accuracy in Tracking Fast Movements

Shreya Saxena, Sridevi V. Sarma, Munther A. Dahleh
https://doi.org/10.1162/neco_a_01272 | pp. 865-886

The ability to move fast and accurately track moving objects is fundamentally constrained by the biophysics of neurons and dynamics of the muscles involved. Yet the corresponding trade-offs between these factors and tracking motor commands have not been rigorously quantified. We use feedback control principles to quantify performance limitations of the sensorimotor control system (SCS) to track fast periodic movements. We show that (1) linear models of the SCS fail to predict known undesirable phenomena, including skipped cycles, overshoot and undershoot, produced when tracking signals in the "fast regime," while nonlinear pulsatile control models can predict such undesirable phenomena, and (2) tools from nonlinear control theory allow us to characterize fundamental limitations in this fast regime. Using a validated and tractable nonlinear model of the SCS, we derive an analytical upper bound on frequencies that the SCS model can reliably track before producing such undesirable phenomena as a function of the neurons' biophysical constraints and muscle dynamics. The performance limitations derived here have important implications in sensorimotor control. For example, if the primary motor cortex is compromised due to disease or damage, the theory suggests ways to manipulate muscle dynamics by adding the necessary compensatory forces using an assistive neuroprosthetic device to restore motor performance and, more important, fast and agile movements. Just how one should compensate can be informed by our SCS model and the theory developed here.

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27. Comparison of Different Spike Train Synchrony Measures Regarding Their Robustness to Erroneous Data From Bicuculline-Induced Epileptiform Activity

Manuel Ciba, Robert Bestel, Christoph Nick, Guilherme Ferraz de Arruda, Thomas Peron, Comin César Henrique, Luciano da Fontoura Costa, Francisco Aparecido Rodrigues, Christiane Thielemann
https://doi.org/10.1162/neco_a_01277 | pp. 887-911

As synchronized activity is associated with basic brain functions and pathological states, spike train synchrony has become an important measure to analyze experimental neuronal data. Many measures of spike train synchrony have been proposed, but there is no gold standard allowing for comparison of results from different experiments. This work aims to provide guidance on which synchrony measure is best suited to quantify the effect of epileptiform-inducing substances (e.g., bicuculline, BIC) in in vitro neuronal spike train data. Spike train data from recordings are likely to suffer from erroneous spike detection, such as missed spikes (false negative) or noise (false positive). Therefore, different timescale-dependent (cross-correlation, mutual information, spike time tiling coefficient) and timescale-independent (Spike-contrast, phase synchronization (PS), A-SPIKE-synchronization, A-ISI-distance, ARI-SPIKE-distance) synchrony measures were compared in terms of their robustness to erroneous spike trains. For this purpose, erroneous spike trains were generated by randomly adding (false positive) or deleting (false negative) spikes (in silico manipulated data) from experimental data. In addition, experimental data were analyzed using different spike detection threshold factors in order to confirm the robustness of the synchrony measures. All experimental data were recorded from cortical neuronal networks on microelectrode array chips, which show epileptiform activity induced by the substance BIC. As a result of the in silico manipulated data, Spike-contrast was the only measure that was robust to false-negative as well as false-positive spikes. Analyzing the experimental data set revealed that all measures were able to capture

the effect of BIC in a statistically significant way, with Spike-contrast showing the highest statistical significance even at low spike detection thresholds. In summary, we suggest using Spike-contrast to complement established synchrony measures because it is timescale independent and robust to erroneous spike trains.

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28. Equivalence Projective Simulation as a Framework for Modeling Formation of Stimulus Equivalence Classes

Asieh Abolpour Mofrad, Anis Yazidi, Hugo L. Hammer, Erik Arntzen

https://doi.org/10.1162/neco_a_01274 | pp. 912-968

Stimulus equivalence (SE) and projective simulation (PS) study complex behavior, the former in human subjects and the latter in artificial agents. We apply the PS learning framework for modeling the formation of equivalence classes. For this purpose, we first modify the PS model to accommodate imitating the emergence of equivalence relations. Later, we formulate the SE formation through the matching-to-sample (MTS) procedure. The proposed version of PS model, called the equivalence projective simulation (EPS) model, is able to act within a varying action set and derive new relations without receiving feedback from the environment. To the best of our knowledge, it is the first time that the field of equivalence theory in behavior analysis has been linked to an artificial agent in a machine learning context. This model has many advantages over existing neural network models. Briefly, our EPS model is not a black box model, but rather a model with the capability of easy interpretation and flexibility for further modifications. To validate the model, some experimental results performed by prominent behavior analysts are simulated. The results confirm that the EPS model is able to reliably simulate and replicate the same behavior as real experiments in various settings, including formation of equivalence relations in typical participants, nonformation of equivalence relations in language-disabled children, and nodal effect in a linear series with nodal distance five. Moreover, through a hypothetical experiment, we discuss the possibility of applying EPS in further equivalence theory research.

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29. The Discriminative Kalman Filter for Bayesian Filtering with Nonlinear and Nongaussian Observation Models

Michael C. Burkhart, David M. Brandman, Brian Franco, Leigh R. Hochberg, Matthew Tom Harrison

https://doi.org/10.1162/neco_a_01275 | pp. 969-1017

Abstract The Kalman filter provides a simple and efficient algorithm to compute the posterior distribution for state-space models where both the latent state and measurement models are linear and gaussian. Extensions to the Kalman filter, including the extended and unscented Kalman filters, incorporate linearizations for models where the observation model p(observation|state) is nonlinear. We argue that in many cases, a model for p(state|observation) proves both easier to learn and more accurate for latent state estimation. Approximating p(state|observation) as gaussian leads to a new filtering algorithm, the discriminative Kalman filter (DKF), which can perform well even when p(observation|state) is highly nonlinear and/or nongaussian. The approximation, motivated by the Bernstein–von Mises theorem, improves as the dimensionality of the observations increases. The DKF has computational complexity similar to the Kalman filter, allowing it in some cases to perform much faster than particle filters with similar precision, while better accounting for nonlinear and nongaussian observation models than Kalman-based

extensions. When the observation model must be learned from training data prior to filtering, off-the-shelf nonlinear and nonparametric regression techniques can provide a gaussian model for p(observation|state) that cleanly integrates with the DKF. As part of the BrainGate2 clinical trial, we successfully implemented gaussian process regression with the DKF framework in a brain-computer interface to provide real-time, closed-loop cursor control to a person with a complete spinal cord injury. In this letter, we explore the theory underlying the DKF, exhibit some illustrative examples, and outline potential extensions.

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30. The Stochastic Delta Rule: Faster and More Accurate Deep Learning Through Adaptive Weight Noise

Noah Frazier-Logue, Stephen José Hanson

https://doi.org/10.1162/neco_a_01276 | pp. 1018-1032

Abstract Multilayer neural networks have led to remarkable performance on many kinds of benchmark tasks in text, speech, and image processing. Nonlinear parameter estimation in hierarchical models is known to be subject to overfitting and misspecification. One approach to these estimation and related problems (e.g., saddle points, colinearity, feature discovery) is called Dropout. The Dropout algorithm removes hidden units according to a binomial random variable with probability p prior to each update, creating random “shocks” to the network that are averaged over updates (thus creating weight sharing). In this letter, we reestablish an older parameter search method and show that Dropout is a special case of this more general model, stochastic delta rule (SDR), published originally in 1990. Unlike Dropout, SDR redefines each weight in the network as a random variable with mean μwij and standard deviation σwij. Each weight random variable is sampled on each forward activation, consequently creating an exponential number of potential networks with shared weights (accumulated in the mean values). Both parameters are updated according to prediction error, thus resulting in weight noise injections that reflect a local history of prediction error and local model averaging. SDR therefore implements a more sensitive local gradient-dependent simulated annealing per weight converging in the limit to a Bayes optimal network. We run tests on standard benchmarks (CIFAR and ImageNet) using a modified version of DenseNet and show that SDR outperforms standard Dropout in top-5 validation error by approximately 13% with DenseNet-BC 121 on ImageNet and find various validation error improvements in smaller networks. We also show that SDR reaches the same accuracy that Dropout attains in 100 epochs in as few as 40 epochs, as well as improvements in training error by as much as 80%.

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31. Nonequilibrium Statistical Mechanics of Continuous Attractors

Weishun Zhong, Zhiyue Lu, David J. Schwab, Arvind Murugan

https://doi.org/10.1162/neco_a_01280 | pp. 1033-1068

Continuous attractors have been used to understand recent neuroscience experiments where persistent activity patterns encode internal representations of external attributes like head direction or spatial location. However, the conditions under which the emergent bump of neural activity in such networks can be manipulated by space and time-dependent external sensory or motor signals are not understood. Here, we find fundamental limits on how rapidly internal representations encoded along continuous attractors can be updated by an external signal. We apply these results to place cell networks to derive a velocity-dependent nonequilibrium memory capacity in neural networks.

32. First Passage Time Memory Lifetimes for Multistate, Filter-Based Synapses

Terry Elliott

https://doi.org/10.1162/neco_a_01283 | pp. 1069-1143

Models of associative memory with discrete state synapses learn new memories by forgetting old ones. In contrast to non-integrative models of synaptic plasticity, models with integrative, filter-based synapses exhibit an initial rise in the fidelity of recall of stored memories. This rise to a peak is driven by a transient process and is then followed by a return to equilibrium. In a series of papers, we have employed a first passage time (FPT) approach to define and study memory lifetimes, incrementally developing our methods, from both simple and complex binary-strength synapses to simple multistate synapses. Here, we complete this work by analyzing FPT memory lifetimes in multistate, filter-based synapses. To achieve this, we integrate out the internal filter states so that we can work with transitions only in synaptic strength. We then generalize results on polysynaptic generating functions from binary strength to multistate synapses, allowing us to examine the dynamics of synaptic strength changes in an ensemble of synapses rather than just a single synapse. To derive analytical results for FPT memory lifetimes, we partition the synaptic dynamics into two distinct phases: the first, pre-peak phase studied with a drift-only approximation, and the second, post-peak phase studied with approximations to the full strength transition probabilities. These approximations capture the underlying dynamics very well, as demonstrated by the extremely good agreement between results obtained by simulating our model and results obtained from the Fokker-Planck or integral equation approaches to FPT processes.

33. Efficient Position Decoding Methods Based on Fluorescence Calcium Imaging in the Mouse Hippocampus

Mengyu Tu, Ruohe Zhao, Avital Adler, Wen■Biao Gan, Zhe Chen

https://doi.org/10.1162/neco_a_01281 | pp. 1144-1167

Large-scale fluorescence calcium imaging methods have become widely adopted for studies of long-term hippocampal and cortical neuronal dynamics. Pyramidal neurons of the rodent hippocampus show spatial tuning in freely foraging or head-fixed navigation tasks. Development of efficient neural decoding methods for reconstructing the animal's position in real or virtual environments can provide a fast readout of spatial representations in closed-loop neuroscience experiments. Here, we develop an efficient strategy to extract features from fluorescence calcium imaging traces and further decode the animal's position. We validate our spike inference-free decoding methods in multiple in vivo calcium imaging recordings of the mouse hippocampus based on both supervised and unsupervised decoding analyses. We systematically investigate the decoding performance of our proposed methods with respect to the number of neurons, imaging frame rate, and signal-to-noise ratio. Our proposed supervised decoding analysis is ultrafast and robust, and thereby appealing for real-time position decoding applications based on calcium imaging.

34. Independently Interpretable Lasso for Generalized Linear Models

Masaaki Takada, Taiji Suzuki, Hironori Fujisawa

https://doi.org/10.1162/neco_a_01279 | pp. 1168-1221

Sparse regularization such as [Formula: see text] regularization is a quite powerful and widely used strategy for high-dimensional learning problems. The effectiveness of sparse regularization has been supported practically and theoretically by several studies. However, one of the biggest issues in sparse regularization is that its performance is quite sensitive to correlations between features. Ordinary [Formula: see text] regularization selects variables correlated with each other under weak regularizations, which results in deterioration of not only its estimation error but also interpretability. In this letter, we propose a new regularization method, independently interpretable lasso (IILasso), for generalized linear models. Our proposed regularizer suppresses selecting correlated variables, so that each active variable affects the response independently in the model. Hence, we can interpret regression coefficients intuitively, and the performance is also improved by avoiding overfitting. We analyze the theoretical property of the IILasso and show that the proposed method is advantageous for its sign recovery and achieves almost minimax optimal convergence rate. Synthetic and real data analyses also indicate the effectiveness of the IILasso.

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35. Salient Slices: Improved Neural Network Training and Performance with Image Entropy

Steven J. Frank, Andrea Frank

https://doi.org/10.1162/neco_a_01282 | pp. 1222-1237

As a training and analysis strategy for convolutional neural networks (CNNs), we slice images into tiled segments and use, for training and prediction, segments that both satisfy an information criterion and contain sufficient content to support classification. In particular, we use image entropy as the information criterion. This ensures that each tile carries as much information diversity as the original image and, for many applications, serves as an indicator of usefulness in classification. To make predictions, a probability aggregation framework is applied to probabilities assigned by the CNN to the input image tiles. This technique, which we call Salient Slices, facilitates the use of large, high-resolution images that would be impractical to analyze unmodified; provides data augmentation for training, which is particularly valuable when image availability is limited; and the ensemble nature of the input for prediction enhances its accuracy.

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36. Heterogeneous Synaptic Weighting Improves Neural Coding in the Presence of Common Noise

Pratik Sachdeva, Jesse A. Livezey, Michael R. DeWeese

https://doi.org/10.1162/neco_a_01287 | pp. 1239-1276

Simultaneous recordings from the cortex have revealed that neural activity is highly variable and that some variability is shared across neurons in a population. Further experimental work has demonstrated that the shared component of a neuronal population's variability is typically comparable to or larger than its private component. Meanwhile, an abundance of theoretical work has assessed the impact that shared variability has on a population code. For example, shared input noise is understood to have a detrimental impact on a neural population's coding fidelity. However, other contributions to variability, such as common noise, can also play a role in shaping correlated variability. We present a network of linear-nonlinear neurons in which we introduce a common noise input to model-for instance, variability resulting from upstream action

potentials that are irrelevant to the task at hand. We show that by applying a heterogeneous set of synaptic weights to the neural inputs carrying the common noise, the network can improve its coding ability as measured by both Fisher information and Shannon mutual information, even in cases where this results in amplification of the common noise. With a broad and heterogeneous distribution of synaptic weights, a population of neurons can remove the harmful effects imposed by afferents that are uninformative about a stimulus. We demonstrate that some nonlinear networks benefit from weight diversification up to a certain population size, above which the drawbacks from amplified noise dominate over the benefits of diversification. We further characterize these benefits in terms of the relative strength of shared and private variability sources. Finally, we studied the asymptotic behavior of the mutual information and Fisher information analytically in our various networks as a function of population size. We find some surprising qualitative changes in the asymptotic behavior as we make seemingly minor changes in the synaptic weight distributions.

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37. A Model for the Study of the Increase in Stimulus and Change Point Detection with Small and Variable Spiking Delays

Benjamin Straub, Gaby Schneider

https://doi.org/10.1162/neco_a_01285 | pp. 1277-1321

Precise timing of spikes between different neurons has been found to convey reliable information beyond the spike count. In contrast, the role of small and variable spiking delays, as reported, for example, in the visual cortex, remains largely unclear. This issue becomes particularly important considering the high speed of neuronal information processing, which is assumed to be based on only a few milliseconds within each processing step. We investigate the role of small and variable spiking delays with a parsimonious stochastic spiking model that is strongly motivated by experimental observations. The model contains only two parameters for the response of a neuron to one stimulus, describing directly the rate and the delay, or phase. Within the theoretical model, we specifically investigate two quantities, the probability of correct stimulus detection and the probability of correct change point detection, as a function of these parameters and within short periods of time. Optimal combinations of the two parameters across stimuli are derived that maximize these probabilities and enable comparison of pure rate, pure phase, and combined codes. In particular, the gain in correct detection probability when adding small and variable spiking delays to pure rate coding increases with the number of stimuli. More interesting, small and variable spiking delays can considerably improve the process of detecting changes in the stimulus, while also decreasing the probability of false alarms and thus increasing robustness and speed of change point detection. The results are compared to empirical spike train recordings of neurons in the visual cortex reported earlier in response to a number of visual stimuli. The results suggest that near-optimal combinations of rate and phase parameters may be implemented in the brain and that adding phase information could particularly increase the quality of change point detection in cases of highly similar stimuli.

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38. A Mathematical Analysis of Memory Lifetime in a Simple Network Model of Memory

Pascal Helson

https://doi.org/10.1162/neco_a_01286 | pp. 1322-1354

We study the learning of an external signal by a neural network and the time to forget it when this network is submitted to noise. The presentation of an external stimulus to the recurrent network of binary neurons may change the state of the synapses. Multiple presentations of a unique signal lead to its learning. Then, during the forgetting time, the presentation of other signals (noise) may also modify the synaptic weights. We construct an estimator of the initial signal using the synaptic currents and in this way define a probability of error. In our model, these synaptic currents evolve as Markov chains. We study the dynamics of these Markov chains and obtain a lower bound on the number of external stimuli that the network can receive before the initial signal is considered forgotten (probability of error above a given threshold). Our results are based on a finite-time analysis rather than large-time asymptotic. We finally present numerical illustrations of our results.

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39. Shapley Homology: Topological Analysis of Sample Influence for Neural Networks

Kaixuan Zhang, Qinglong Wang, Xue Liu, C. Lee Giles

https://doi.org/10.1162/neco_a_01289 | pp. 1355-1378

Data samples collected for training machine learning models are typically assumed to be independent and identically distributed (i.i.d.). Recent research has demonstrated that this assumption can be problematic as it simplifies the manifold of structured data. This has motivated different research areas such as data poisoning, model improvement, and explanation of machine learning models. In this work, we study the influence of a sample on determining the intrinsic topological features of its underlying manifold. We propose the Shapley homology framework, which provides a quantitative metric for the influence of a sample of the homology of a simplicial complex. Our proposed framework consists of two main parts: homology analysis, where we compute the Betti number of the target topological space, and Shapley value calculation, where we decompose the topological features of a complex built from data points to individual points. By interpreting the influence as a probability measure, we further define an entropy that reflects the complexity of the data manifold. Furthermore, we provide a preliminary discussion of the connection of the Shapley homology to the Vapnik-Chervonenkis dimension. Empirical studies show that when the zero-dimensional Shapley homology is used on neighboring graphs, samples with higher influence scores have a greater impact on the accuracy of neural networks that determine graph connectivity and on several regular grammars whose higher entropy values imply greater difficulty in being learned.

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40. Generation of Scale-Invariant Sequential Activity in Linear Recurrent Networks

Yue Liu, Marc W. Howard

https://doi.org/10.1162/neco_a_01288 | pp. 1379-1407

Sequential neural activity has been observed in many parts of the brain and has been proposed as a neural mechanism for memory. The natural world expresses temporal relationships at a wide range of scales. Because we cannot know the relevant scales a priori, it is desirable that memory, and thus the generated sequences, is scale invariant. Although recurrent neural network models have been proposed as a mechanism for generating sequences, the requirements for scale-invariant sequences are not known. This letter reports the constraints that enable a linear recurrent neural network model to generate scale-invariant

sequential activity. A straightforward eigendecomposition analysis results in two independent conditions that are required for scale invariance for connectivity matrices with real, distinct eigenvalues. First, the eigenvalues of the network must be geometrically spaced. Second, the eigenvectors must be related to one another via translation. These constraints are easily generalizable for matrices that have complex and distinct eigenvalues. Analogous albeit less compact constraints hold for matrices with degenerate eigenvalues. These constraints, along with considerations on initial conditions, provide a general recipe to build linear recurrent neural networks that support scale-invariant sequential activity.

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41. Minimal Spiking Neuron for Solving Multilabel Classification Tasks

Jakub Fil, Dominique Chu

https://doi.org/10.1162/neco_a_01290 | pp. 1408-1429

The multispike tempotron (MST) is a powersul, single spiking neuron model that can solve complex supervised classification tasks. It is also internally complex, computationally expensive to evaluate, and unsuitable for neuromorphic hardware. Here we aim to understand whether it is possible to simplify the MST model while retaining its ability to learn and process information. To this end, we introduce a family of generalized neuron models (GNMs) that are a special case of the spike response model and much simpler and cheaper to simulate than the MST. We find that over a wide range of parameters, the GNM can learn at least as well as the MST does. We identify the temporal autocorrelation of the membrane potential as the most important ingredient of the GNM that enables it to classify multiple spatiotemporal patterns. We also interpret the GNM as a chemical system, thus conceptually bridging computation by neural networks with molecular information processing. We conclude the letter by proposing alternative training approaches for the GNM, including error trace learning and error backpropagation.

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42. Any Target Function Exists in a Neighborhood of Any Sufficiently Wide Random Network: A Geometrical Perspective

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https://doi.org/10.1162/neco_a_01295 | pp. 1431-1447

It is known that any target function is realized in a sufficiently small neighborhood of any randomly connected deep network, provided the width (the number of neurons in a layer) is sufficiently large. There are sophisticated analytical theories and discussions concerning this striking fact, but rigorous theories are very complicated. We give an elementary geometrical proof by using a simple model for the purpose of elucidating its structure. We show that high-dimensional geometry plays a magical role. When we project a high-dimensional sphere of radius 1 to a low-dimensional subspace, the uniform distribution over the sphere shrinks to a gaussian distribution with negligibly small variances and covariances.

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43. Inference of a Mesoscopic Population Model from Population Spike Trains

Alexandre René, André Longtin, Jakob H. Macke

https://doi.org/10.1162/neco_a_01292 | pp. 1448-1498

Understanding how rich dynamics emerge in neural populations requires models exhibiting a wide range of behaviors while remaining interpretable in terms of connectivity and single-neuron dynamics. However, it has been challenging to fit such mechanistic spiking networks at the single-neuron scale to empirical population data. To close this gap, we propose to fit such data at a mesoscale, using a mechanistic but low-dimensional and, hence, statistically tractable model. The mesoscopic representation is obtained by approximating a population of neurons as multiple homogeneous pools of neurons and modeling the dynamics of the aggregate population activity within each pool. We derive the likelihood of both single-neuron and connectivity parameters given this activity, which can then be used to optimize parameters by gradient ascent on the log likelihood or perform Bayesian inference using Markov chain Monte Carlo (MCMC) sampling. We illustrate this approach using a model of generalized integrate-and-fire neurons for which mesoscopic dynamics have been previously derived and show that both single-neuron and connectivity parameters can be recovered from simulated data. In particular, our inference method extracts posterior correlations between model parameters, which define parameter subsets able to reproduce the data. We compute the Bayesian posterior for combinations of parameters using MCMC sampling and investigate how the approximations inherent in a mesoscopic population model affect the accuracy of the inferred single-neuron parameters.

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44. Stochastic Multichannel Ranking with Brain Dynamics Preferences

Yuangang Pan, Ivor W. Tsang, Avinash Kumar Singh, Chin■Teng Lin, Masashi Sugiyama

https://doi.org/10.1162/neco_a_01293 | pp. 1499-1530

A driver's cognitive state of mental fatigue significantly affects his or her driving performance and more important, public safety. Previous studies have leveraged reaction time (RT) as the metric for mental fatigue and aim at estimating the exact value of RT using electroencephalogram (EEG) signals within a regression model. However, due to the easily corrupted and also nonsmooth properties of RTs during data collection, methods focusing on predicting the exact value of a noisy measurement, RT generally suffer from poor generalization performance. Considering that human RT is the reflection of brain dynamics preference (BDP) rather than a single regression output of EEG signals, we propose a novel channel-reliability-aware ranking (CArank) model for the multichannel ranking problem. CArank learns from BDPs using EEG data robustly and aims at preserving the ordering corresponding to RTs. In particular, we introduce a transition matrix to characterize the reliability of each channel used in the EEG data, which helps in learning with BDPs only from informative EEG channels. To handle large-scale EEG signals, we propose a stochastic-generalized expectation maximum (SGEM) algorithm to update CArank in an online fashion. Comprehensive empirical analysis on EEG signals from 40 participants shows that our CArank achieves substantial improvements in reliability while simultaneously detecting noisy or less informative EEG channels.

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45. A Discrete-Time Neurodynamic Approach to Sparsity-Constrained Nonnegative Matrix Factorization

Xinqi Li, Jun Wang, Sam Kwong

https://doi.org/10.1162/neco_a_01294 | pp. 1531-1562

Sparsity is a desirable property in many nonnegative matrix factorization (NMF) applications. Although some level of sparseness of NMF solutions can be achieved by using regularization, the resulting sparsity depends highly on the regularization parameter to be valued in an ad hoc way. In this letter we formulate sparse NMF as a mixed-integer optimization problem with sparsity as binary constraints. A discrete-time projection neural network is developed for solving the formulated problem. Sufficient conditions for its stability and convergence are analytically characterized by using Lyapunov's method. Experimental results on sparse feature extraction are discussed to substantiate the superiority of this approach to extracting highly sparse features.

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46. On a Scalable Entropic Breaching of the Overfitting Barrier for Small Data Problems in Machine Learning

Illia Horenko

https://doi.org/10.1162/neco_a_01296 | pp. 1563-1579

Overfitting and treatment of small data are among the most challenging problems in machine learning (ML), when a relatively small data statistics size [Formula: see text] is not enough to provide a robust ML fit for a relatively large data feature dimension [Formula: see text]. Deploying a massively parallel ML analysis of generic classification problems for different [Formula: see text] and [Formula: see text], we demonstrate the existence of statistically significant linear overfitting barriers for common ML methods. The results reveal that for a robust classification of bioinformatics-motivated generic problems with the long short-term memory deep learning classifier (LSTM), one needs in the best case a statistics [Formula: see text] that is at least 13.8 times larger than the feature dimension [Formula: see text]. We show that this overfitting barrier can be breached at a 10[Formula: see text] fraction of the computational cost by means of the entropy-optimal scalable probabilistic approximations algorithm (eSPA), performing a joint solution of the entropy-optimal Bayesian network inference and feature space segmentation problems. Application of eSPA to experimental single cell RNA sequencing data exhibits a 30-fold classification performance boost when compared to standard bioinformatics tools and a 7-fold boost when compared to the deep learning LSTM classifier.

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47. Theory and Algorithms for Shapelet-Based Multiple-Instance Learning

Daiki Suehiro, Kohei Hatano, Eiji Takimoto, Shuji Yamamoto, Kenichi Bannai, Akiko Takeda

https://doi.org/10.1162/neco_a_01297 | pp. 1580-1613

We propose a new formulation of multiple-instance learning (MIL), in which a unit of data consists of a set of instances called a bag. The goal is to find a good classifier of bags based on the similarity with a "shapelet" (or pattern), where the similarity of a bag with a shapelet is the maximum similarity of instances in the bag. In previous work, some of the training instances have been chosen as shapelets with no theoretical justification. In our formulation, we use all possible, and thus infinitely many, shapelets, resulting in a richer class of classifiers. We show that the formulation is tractable, that is, it can be reduced through linear programming boosting (LPBoost) to difference of convex (DC) programs of finite (actually polynomial) size. Our theoretical result also gives justification to the heuristics of some previous work. The time complexity of the proposed algorithm highly depends on the size of the set of all instances in the training sample. To apply to the data containing a large number of instances, we also propose a heuristic option of the algorithm without the loss of the theoretical guarantee. Our empirical study demonstrates that our algorithm uniformly

works for shapelet learning tasks on time-series classification and various MIL tasks with comparable accuracy to the existing methods. Moreover, we show that the proposed heuristics allow us to achieve the result in reasonable computational time.

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48. A Mean-Field Description of Bursting Dynamics in Spiking Neural Networks with Short-Term Adaptation

Richard Gast, Helmut Schmidt, Thomas R. Knösche

https://doi.org/10.1162/neco_a_01300 | pp. 1615-1634

Bursting plays an important role in neural communication. At the population level, macroscopic bursting has been identified in populations of neurons that do not express intrinsic bursting mechanisms. For the analysis of phase transitions between bursting and non-bursting states, mean-field descriptions of macroscopic bursting behavior are a valuable tool. In this article, we derive mean-field descriptions of populations of spiking neurons and examine whether states of collective bursting behavior can arise from short-term adaptation mechanisms. Specifically, we consider synaptic depression and spike-frequency adaptation in networks of quadratic integrate-and-fire neurons. Analyzing the mean-field model via bifurcation analysis, we find that bursting behavior emerges for both types of short-term adaptation. This bursting behavior can coexist with steady-state behavior, providing a bistable regime that allows for transient switches between synchronized and nonsynchronized states of population dynamics. For all of these findings, we demonstrate a close correspondence between the spiking neural network and the mean-field model. Although the mean-field model has been derived under the assumptions of an infinite population size and all-to-all coupling inside the population, we show that this correspondence holds even for small, sparsely coupled networks. In summary, we provide mechanistic descriptions of phase transitions between bursting and steady-state population dynamics, which play important roles in both healthy neural communication and neurological disorders.

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49. Parallel Neural Multiprocessing with Gamma Frequency Latencies

Ruohan Zhang, Dana H. Ballard

https://doi.org/10.1162/neco_a_01301 | pp. 1635-1663

The Poisson variability in cortical neural responses has been typically modeled using spike averaging techniques, such as trial averaging and rate coding, since such methods can produce reliable correlates of behavior. However, mechanisms that rely on counting spikes could be slow and inefficient and thus might not be useful in the brain for computations at timescales in the 10 millisecond range. This issue has motivated a search for alternative spike codes that take advantage of spike timing and has resulted in many studies that use synchronized neural networks for communication. Here we focus on recent studies that suggest that the gamma frequency may provide a reference that allows local spike phase representations that could result in much faster information transmission. We have developed a unified model (gamma spike multiplexing) that takes advantage of a single cycle of a cell's somatic gamma frequency to modulate the generation of its action potentials. An important consequence of this coding mechanism is that it allows multiple independent neural processes to run in parallel, thereby greatly increasing the processing capability of the cortex. System-level simulations and preliminary analysis of mouse cortical cell data are presented as support for the proposed theoretical model.

50. Fine-Grained 3D-Attention Prototypes for Few-Shot Learning

Xin Hu, Jun Liu, Jie Ma, Yudai Pan, Lingling Zhang

https://doi.org/10.1162/neco_a_01302 | pp. 1664-1684

In the real world, a limited number of labeled finely grained images per class can hardly represent the class distribution effectively. Due to the more subtle visual differences in fine-grained images than simple images with obvious objects, that is, there exist smaller interclass and larger intraclass variations. To solve these issues, we propose an end-to-end attention-based model for fine-grained few-shot image classification (AFG) with the recent episode training strategy. It is composed mainly of a feature learning module, an image reconstruction module, and a label distribution module. The feature learning module mainly devises a 3D-Attention mechanism, which considers both the spatial positions and different channel attentions of the image features, in order to learn more discriminative local features to better represent the class distribution. The image reconstruction module calculates the mappings between local features and the original images. It is constrained by a designed loss function as auxiliary supervised information, so that the learning of each local feature does not need extra annotations. The label distribution module is used to predict the label distribution of a given unlabeled sample, and we use the local features to represent the image features for classification. By conducting comprehensive experiments on Mini-ImageNet and three fine-grained data sets, we demonstrate that the proposed model achieves superior performance over the competitors.

51. Hyperbolic-Valued Hopfield Neural Networks in Synchronous Mode

Masaki Kobayashi

https://doi.org/10.1162/neco_a_01303 | pp. 1685-1696

For most multistate Hopfield neural networks, the stability conditions in asynchronous mode are known, whereas those in synchronous mode are not. If they were to converge in synchronous mode, recall would be accelerated by parallel processing. Complex-valued Hopfield neural networks (CHNNs) with a projection rule do not converge in synchronous mode. In this work, we provide stability conditions for hyperbolic Hopfield neural networks (HHNNs) in synchronous mode instead of CHNNs. HHNNs provide better noise tolerance than CHNNs. In addition, the stability conditions are applied to the projection rule, and HHNNs with a projection rule converge in synchronous mode. By computer simulations, we find that the projection rule for HHNNs in synchronous mode maintains a high noise tolerance.

52. Tensor Least Angle Regression for Sparse Representations of Multidimensional Signals

Ishan Wickramasingha, Ahmed Elrewainy, Michael Sobhy, Sherif S. Sherif

https://doi.org/10.1162/neco_a_01304 | pp. 1697-1732

Sparse signal representations have gained much interest recently in both signal processing and statistical communities. Compared to orthogonal matching pursuit (OMP) and basis pursuit, which solve the [Formula: see text] and [Formula: see text] constrained sparse least-squares problems, respectively, least angle regression (LARS) is a computationally efficient method to solve both problems for all critical values of the regularization parameter [Formula: see text]. However, all of these methods are not suitable for solving large

multidimensional sparse least-squares problems, as they would require extensive computational power and memory. An earlier generalization of OMP, known as Kronecker-OMP, was developed to solve the [Formula: see text] problem for large multidimensional sparse least-squares problems. However, its memory usage and computation time increase quickly with the number of problem dimensions and iterations. In this letter, we develop a generalization of LARS, tensor least angle regression (T-LARS) that could efficiently solve either large [Formula: see text] or large [Formula: see text] constrained multidimensional, sparse, least-squares problems (underdetermined or overdetermined) for all critical values of the regularization parameter [Formula: see text] and with lower computational complexity and memory usage than Kronecker-OMP. To demonstrate the validity and performance of our T-LARS algorithm, we used it to successfully obtain different sparse representations of two relatively large 3D brain images, using fixed and learned separable overcomplete dictionaries, by solving both [Formula: see text] and [Formula: see text] constrained sparse least-squares problems. Our numerical experiments demonstrate that our T-LARS algorithm is significantly faster (46 to 70 times) than Kronecker-OMP in obtaining [Formula: see text]-sparse solutions for multilinear leastsquares problems. However, the [Formula: see text]-sparse solutions obtained using Kronecker-OMP always have a slightly lower residual error (1.55% to 2.25%) than ones obtained by T-LARS. Therefore, T-LARS could be an important tool for numerous multidimensional biomedical signal processing applications.

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53. Polynomial-Time Algorithms for Multiple-Arm Identification with Full-Bandit Feedback

Yuko Kuroki, Liyuan Xu, Atsushi Miyauchi, Junya Honda, Masashi Sugiyama

https://doi.org/10.1162/neco_a_01299 | pp. 1733-1773

We study the problem of stochastic multiple-arm identification, where an agent sequentially explores a size-[Formula: see text] subset of arms (also known as a super arm) from given [Formula: see text] arms and tries to identify the best super arm. Most work so far has considered the semi-bandit setting, where the agent can observe the reward of each pulled arm or assumed each arm can be queried at each round. However, in real-world applications, it is costly or sometimes impossible to observe a reward of individual arms. In this study, we tackle the full-bandit setting, where only a noisy observation of the total sum of a super arm is given at each pull. Although our problem can be regarded as an instance of the best arm identification in linear bandits, a naive approach based on linear bandits is computationally infeasible since the number of super arms [Formula: see text] is exponential. To cope with this problem, we first design a polynomial-time approximation algorithm for a 0-1 quadratic programming problem arising in confidence ellipsoid maximization. Based on our approximation algorithm, we propose a bandit algorithm whose computation time is [Formula: see text](log [Formula: see text]), thereby achieving an exponential speedup over linear bandit algorithms. We provide a sample complexity upper bound that is still worst-case optimal. Finally, we conduct experiments on large-scale data sets with more than 10[Formula: see text] super arms, demonstrating the superiority of our algorithms in terms of both the computation time and the sample complexity.

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54. Fast and Accurate Langevin Simulations of Stochastic Hodgkin-Huxley Dynamics

Shusen Pu, Peter J. Thomas
https://doi.org/10.1162/neco_a_01312 | pp. 1775-1835

Fox and Lu introduced a Langevin framework for discrete-time stochastic models of randomly gated ion channels such as the Hodgkin-Huxley (HH) system. They derived a Fokker-Planck equation with state-dependent diffusion tensor [Formula: see text] and suggested a Langevin formulation with noise coefficient matrix [Formula: see text] such that SS[Formula: see text]. Subsequently, several authors introduced a variety of Langevin equations for the HH system. In this article, we present a natural 14-dimensional dynamics for the HH system in which each directed edge in the ion channel state transition graph acts as an independent noise source, leading to a 14 [Formula: see text] 28 noise coefficient matrix [Formula: see text]. We show that (1) the corresponding 14D system of ordinary differential equations is consistent with the classical 4D representation of the HH system; (2) the 14D representation leads to a noise coefficient matrix [Formula: see text] that can be obtained cheaply on each time step, without requiring a matrix decomposition; (3) sample trajectories of the 14D representation are pathwise equivalent to trajectories of Fox and Lu's system, as well as trajectories of several existing Langevin models; (4) our 14D representation (and those equivalent to it) gives the most accurate interspike interval distribution, not only with respect to moments but under both the [Formula: see text] and [Formula: see text] metric-space norms; and (5) the 14D representation gives an approximation to exact Markov chain simulations that are as fast and as efficient as all equivalent models. Our approach goes beyond existing models, in that it supports a stochastic shielding decomposition that dramatically simplifies [Formula: see text] with minimal loss of accuracy under both voltage- and current-clamp conditions.

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55. A Predictive-Coding Network That Is Both Discriminative and Generative

Wei Sun, Jeff Orchard
https://doi.org/10.1162/neco_a_01311 | pp. 1836-1862

Predictive coding (PC) networks are a biologically interesting class of neural networks. Their layered hierarchy mimics the reciprocal connectivity pattern observed in the mammalian cortex, and they can be trained using local learning rules that approximate backpropagation (Bogacz, 2017). However, despite having feedback connections that enable information to flow down the network hierarchy, discriminative PC networks are not typically generative. Clamping the output class and running the network to equilibrium yields an input sample that usually does not resemble the training input. This letter studies this phenomenon and proposes a simple solution that promotes the generation of input samples that resemble the training inputs. Simple decay, a technique already in wide use in neural networks, pushes the PC network toward a unique minimum two-norm solution, and that unique solution provably (for linear networks) matches the training inputs. The method also vastly improves the samples generated for nonlinear networks, as we demonstrate on MNIST.

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56. Binless Kernel Machine: Modeling Spike Train Transformation for Cognitive Neural Prostheses

Cunle Qian, Xuyun Sun, Yueming Wang, Xiaoxiang Zheng, Yiwen Wang, Gang Pan
https://doi.org/10.1162/neco_a_01306 | pp. 1863-1900

Modeling spike train transformation among brain regions helps in designing a cognitive neural prosthesis that restores lost cognitive functions. Various methods analyze the nonlinear dynamic spike train transformation between two cortical areas with low computational eficiency. The application of a real-time neural prosthesis requires computational eficiency, performance stability, and better interpretation of the neural firing patterns that modulate target spike generation. We propose the binless kernel machine in the point-process framework to describe nonlinear dynamic spike train transformations. Our approach embeds the binless kernel to eficiently capture the feedforward dynamics of spike trains and maps the input spike timings into reproducing kernel Hilbert space (RKHS). An inhomogeneous Bernoulli process is designed to combine with a kernel logistic regression that operates on the binless kernel to generate an output spike train as a point process. Weights of the proposed model are estimated by maximizing the log likelihood of output spike trains in RKHS, which allows a global-optimal solution. To reduce computational complexity, we design a streaming-based clustering algorithm to extract typical and important spike train features. The cluster centers and their weights enable the visualization of the important input spike train patterns that motivate or inhibit output neuron firing. We test the proposed model on both synthetic data and real spike train data recorded from the dorsal premotor cortex and the primary motor cortex of a monkey performing a center-out task. Performances are evaluated by discrete-time rescaling Kolmogorov-Smirnov tests. Our model outperforms the existing methods with higher stability regardless of weight initialization and demonstrates higher eficiency in analyzing neural patterns from spike timing with less historical input (50%). Meanwhile, the typical spike train patterns selected according to weights are validated to encode output spike from the spike train of single-input neuron and the interaction of two input neurons.

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57. Modal Principal Component Analysis

Keishi Sando, Hideitsu Hino

https://doi.org/10.1162/neco_a_01308 | pp. 1901-1935

Principal component analysis (PCA) is a widely used method for data processing, such as for dimension reduction and visualization. Standard PCA is known to be sensitive to outliers, and various robust PCA methods have been proposed. It has been shown that the robustness of many statistical methods can be improved using mode estimation instead of mean estimation, because mode estimation is not significantly affected by the presence of outliers. Thus, this study proposes a modal principal component analysis (MPCA), which is a robust PCA method based on mode estimation. The proposed method finds the minor component by estimating the mode of the projected data points. As a theoretical contribution, probabilistic convergence property, influence function, finite-sample breakdown point, and its lower bound for the proposed MPCA are derived. The experimental results show that the proposed method has advantages over conventional methods.

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58. Multiview Alignment and Generation in CCA via Consistent Latent Encoding

Yaxin Shi, Yuangang Pan, Donna Xu, Ivor W. Tsang

https://doi.org/10.1162/neco_a_01309 | pp. 1936-1979

Multiview alignment, achieving one-to-one correspondence of multiview inputs, is critical in many real-world multiview applications, especially for cross-view data analysis problems. An increasing amount of work has

studied this alignment problem with canonical correlation analysis (CCA). However, existing CCA models are prone to misalign the multiple views due to either the neglect of uncertainty or the inconsistent encoding of the multiple views. To tackle these two issues, this letter studies multiview alignment from a Bayesian perspective. Delving into the impairments of inconsistent encodings, we propose to recover correspondence of the multiview inputs by matching the marginalization of the joint distribution of multiview random variables under different forms of factorization. To realize our design, we present adversarial CCA (ACCA), which achieves consistent latent encodings by matching the marginalized latent encodings through the adversarial training paradigm. Our analysis, based on conditional mutual information, reveals that ACCA is flexible for handling implicit distributions. Extensive experiments on correlation analysis and cross-view generation under noisy input settings demonstrate the superiority of our model.

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59. Analysis of Regression Algorithms with Unbounded Sampling

Hongzhi Tong, Jiajing Gao

https://doi.org/10.1162/neco_a_01313 | pp. 1980-1997

In this letter, we study a class of the regularized regression algorithms when the sampling process is unbounded. By choosing different loss functions, the learning algorithms can include a wide range of commonly used algorithms for regression. Unlike the prior work on theoretical analysis of unbounded sampling, no constraint on the output variables is specified in our setting. By an elegant error analysis, we prove consistency and finite sample bounds on the excess risk of the proposed algorithms under regular conditions.

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60. Active Learning of Bayesian Linear Models with High-Dimensional Binary Features by Parameter Confidence-Region Estimation

I. Yu, Masayuki Karasuyama, Keiichi Inoue, Hideki Kandori, Ichiro Takeuchi

https://doi.org/10.1162/neco_a_01310 | pp. 1998-2031

In this letter, we study an active learning problem for maximizing an unknown linear function with high-dimensional binary features. This problem is notoriously complex but arises in many important contexts. When the sampling budget, that is, the number of possible function evaluations, is smaller than the number of dimensions, it tends to be impossible to identify all of the optimal binary features. Therefore, in practice, only a small number of such features are considered, with the majority kept fixed at certain default values, which we call the working set heuristic. The main contribution of this letter is to formally study the working set heuristic and present a suite of theoretically robust algorithms for more efficient use of the sampling budget. Technically, we introduce a novel method for estimating the confidence regions of model parameters that is tailored to active learning with high-dimensional binary features. We provide a rigorous theoretical analysis of these algorithms and prove that a commonly used working set heuristic can identify optimal binary features with favorable sample complexity. We explore the performance of the proposed approach through numerical simulations and an application to a functional protein design problem.

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61. Active Learning for Enumerating Local Minima Based on Gaussian Process Derivatives

I. Yu, Daisuke Sugita, Kazuaki Toyoura, Ichiro Takeuchi

https://doi.org/10.1162/neco_a_01307 | pp. 2032-2068

We study active learning (AL) based on gaussian processes (GPs) for efficiently enumerating all of the local minimum solutions of a black-box function. This problem is challenging because local solutions are characterized by their zero gradient and positive-definite Hessian properties, but those derivatives cannot be directly observed. We propose a new AL method in which the input points are sequentially selected such that the confidence intervals of the GP derivatives are effectively updated for enumerating local minimum solutions. We theoretically analyze the proposed method and demonstrate its usefulness through numerical experiments.

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62. A Cerebellar Computational Mechanism for Delay Conditioning at Precise Time Intervals

Terence D. Sanger, Mitsuo Kawato

https://doi.org/10.1162/neco_a_01318 | pp. 2069-2084

The cerebellum is known to have an important role in sensing and execution of precise time intervals, but the mechanism by which arbitrary time intervals can be recognized and replicated with high precision is unknown. We propose a computational model in which precise time intervals can be identified from the pattern of individual spike activity in a population of parallel fibers in the cerebellar cortex. The model depends on the presence of repeatable sequences of spikes in response to conditioned stimulus input. We emulate granule cells using a population of Izhikevich neuron approximations driven by random but repeatable mossy fiber input. We emulate long-term depression (LTD) and long-term potentiation (LTP) synaptic plasticity at the parallel fiber to Purkinje cell synapse. We simulate a delay conditioning paradigm with a conditioned stimulus (CS) presented to the mossy fibers and an unconditioned stimulus (US) some time later issued to the Purkinje cells as a teaching signal. We show that Purkinje cells rapidly adapt to decrease firing probability following onset of the CS only at the interval for which the US had occurred. We suggest that detection of replicable spike patterns provides an accurate and easily learned timing structure that could be an important mechanism for behaviors that require identification and production of precise time intervals.

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63. Reverse-Engineering Neural Networks to Characterize Their Cost Functions

Takuya Isomura, Karl Friston

https://doi.org/10.1162/neco_a_01315 | pp. 2085-2121

This letter considers a class of biologically plausible cost functions for neural networks, where the same cost function is minimized by both neural activity and plasticity. We show that such cost functions can be cast as a variational bound on model evidence under an implicit generative model. Using generative models based on partially observed Markov decision processes (POMDP), we show that neural activity and plasticity perform Bayesian inference and learning, respectively, by maximizing model evidence. Using mathematical

and numerical analyses, we establish the formal equivalence between neural network cost functions and variational free energy under some prior beliefs about latent states that generate inputs. These prior beliefs are determined by particular constants (e.g., thresholds) that define the cost function. This means that the Bayes optimal encoding of latent or hidden states is achieved when the network's implicit priors match the process that generates its inputs. This equivalence is potentially important because it suggests that any hyperparameter of a neural network can itself be optimized-by minimization with respect to variational free energy. Furthermore, it enables one to characterize a neural network formally, in terms of its prior beliefs.

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64. Closed-Loop Deep Learning: Generating Forward Models With Backpropagation

Sama Daryanavard, Bernd Porr

https://doi.org/10.1162/neco_a_01317 | pp. 2122-2144

A reflex is a simple closed-loop control approach that tries to minimize an error but fails to do so because it will always react too late. An adaptive algorithm can use this error to learn a forward model with the help of predictive cues. For example, a driver learns to improve steering by looking ahead to avoid steering in the last minute. In order to process complex cues such as the road ahead, deep learning is a natural choice. However, this is usually achieved only indirectly by employing deep reinforcement learning having a discrete state space. Here, we show how this can be directly achieved by embedding deep learning into a closed-loop system and preserving its continuous processing. We show in z-space specifically how error backpropagation can be achieved and in general how gradient-based approaches can be analyzed in such closed-loop scenarios. The performance of this learning paradigm is demonstrated using a line follower in simulation and on a real robot that shows very fast and continuous learning.

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65. Assessing Goodness-of-Fit in Marked Point Process Models of Neural Population Coding via Time and Rate Rescaling

Ali Yousefi, Yalda Amidi, Behzad Nazari, Uri T. Eden

https://doi.org/10.1162/neco_a_01321 | pp. 2145-2186

Marked point process models have recently been used to capture the coding properties of neural populations from multiunit electrophysiological recordings without spike sorting. These clusterless models have been shown in some instances to better describe the firing properties of neural populations than collections of receptive field models for sorted neurons and to lead to better decoding results. To assess their quality, we previously proposed a goodness-of-fit technique for marked point process models based on time rescaling, which for a correct model produces a set of uniform samples over a random region of space. However, assessing uniformity over such a region can be challenging, especially in high dimensions. Here, we propose a set of new transformations in both time and the space of spike waveform features, which generate events that are uniformly distributed in the new mark and time spaces. These transformations are scalable to multidimensional mark spaces and provide uniformly distributed samples in hypercubes, which are well suited for uniformity tests. We discuss the properties of these transformations and demonstrate aspects of model fit captured by each transformation. We also compare multiple uniformity tests to determine their power to identify lack-of-fit in the rescaled data. We demonstrate an application of these transformations and uniformity tests in a simulation study. Proofs for each transformation are provided in the

appendix.

66. Inferring Neuronal Couplings From Spiking Data Using a Systematic Procedure With a Statistical Criterion

Yu Terada, Tomoyuki Obuchi, Takuya Isomura, Yoshiyuki Kabashima

https://doi.org/10.1162/neco_a_01324 | pp. 2187-2211

Recent remarkable advances in experimental techniques have provided a background for inferring neuronal couplings from point process data that include a great number of neurons. Here, we propose a systematic procedure for pre- and postprocessing generic point process data in an objective manner to handle data in the framework of a binary simple statistical model, the Ising or generalized McCulloch-Pitts model. The procedure has two steps: (1) determining time bin size for transforming the point process data into discrete-time binary data and (2) screening relevant couplings from the estimated couplings. For the first step, we decide the optimal time bin size by introducing the null hypothesis that all neurons would fire independently, then choosing a time bin size so that the null hypothesis is rejected with the strict criteria. The likelihood associated with the null hypothesis is analytically evaluated and used for the rejection process. For the second postprocessing step, after a certain estimator of coupling is obtained based on the preprocessed data set (any estimator can be used with the proposed procedure), the estimate is compared with many other estimates derived from data sets obtained by randomizing the original data set in the time direction. We accept the original estimate as relevant only if its absolute value is sufficiently larger than those of randomized data sets. These manipulations suppress false positive couplings induced by statistical noise. We apply this inference procedure to spiking data from synthetic and in vitro neuronal networks. The results show that the proposed procedure identifies the presence or absence of synaptic couplings fairly well, including their signs, for the synthetic and experimental data. In particular, the results support that we can infer the physical connections of underlying systems in favorable situations, even when using a simple statistical model.

67. Repetitive Control for Multi-Joint Arm Movements Based on Virtual Trajectories

Yoji Uno, Takehiro Suzuki, Takahiro Kagawa

https://doi.org/10.1162/neco_a_01322 | pp. 2212-2236

According to the neuromuscular model of virtual trajectory control, the postures and movements of limbs are performed by shifting the equilibrium positions determined by agonist and antagonist muscle activities. In this study, we develop virtual trajectory control for the reaching movements of a multi-joint arm, introducing a proportional-derivative feedback control scheme. In virtual trajectory control, it is crucial to design a suitable virtual trajectory such that the desired trajectory can be realized. To this end, we propose an algorithm for updating virtual trajectories in repetitive control, which can be regarded as a Newton-like method in a function space. In our repetitive control, the virtual trajectory is corrected without explicit calculation of the arm dynamics, and the actual trajectory converges to the desired trajectory. Using computer simulations, we assessed the proposed repetitive control for the trajectory tracking of a two-link arm. Our results confirmed that when the feedback gains were reasonably high and the sampling time was sufficiently small, the virtual trajectory was adequately updated, and the desired trajectory was almost achieved within approximately 10

iterative trials. We also propose a method for modifying the virtual trajectory to ensure that the formation of the actual trajectory is identical even when the feedback gains are changed. This modification method makes it possible to execute flexible control, in which the feedback gains are effectively altered according to motion tasks.

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68. Bicomplex Projection Rule for Complex-Valued Hopfield Neural Networks

Masaki Kobayashi

https://doi.org/10.1162/neco_a_01320 | pp. 2237-2248

A complex-valued Hopfield neural network (CHNN) with a multistate activation function is a multistate model of neural associative memory. The weight parameters need a lot of memory resources. Twin-multistate activation functions were introduced to quaternion- and bicomplex-valued Hopfield neural networks. Since their architectures are much more complicated than that of CHNN, the architecture should be simplified. In this work, the number of weight parameters is reduced by bicomplex projection rule for CHNNs, which is given by the decomposition of bicomplex-valued Hopfield neural networks. Computer simulations support that the noise tolerance of CHNN with a bicomplex projection rule is equal to or even better than that of quaternion- and bicomplex-valued Hopfield neural networks. By computer simulations, we find that the projection rule for hyperbolic-valued Hopfield neural networks in synchronous mode maintains a high noise tolerance.

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69. ReLU Networks Are Universal Approximators via Piecewise Linear or Constant Functions

Changcun Huang

https://doi.org/10.1162/neco_a_01316 | pp. 2249-2278

This letter proves that a ReLU network can approximate any continuous function with arbitrary precision by means of piecewise linear or constant approximations. For univariate function [Formula: see text], we use the composite of ReLUs to produce a line segment; all of the subnetworks of line segments comprise a ReLU network, which is a piecewise linear approximation to [Formula: see text]. For multivariate function [Formula: see text], ReLU networks are constructed to approximate a piecewise linear function derived from triangulation methods approximating [Formula: see text]. A neural unit called TRLU is designed by a ReLU network; the piecewise constant approximation, such as Haar wavelets, is implemented by rectifying the linear output of a ReLU network via TRLUs. New interpretations of deep layers, as well as some other results, are also presented.

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70. Effect of Top-Down Connections in Hierarchical Sparse Coding

Victor Boutin, Angelo Franciosini, Franck Ruffier, Laurent Perrinet

https://doi.org/10.1162/neco_a_01325 | pp. 2279-2309

Hierarchical sparse coding (HSC) is a powerful model to efficiently represent multidimensional, structured data such as images. The simplest solution to solve this computationally hard problem is to decompose it into independent layer-wise subproblems. However, neuroscientific evidence would suggest interconnecting

these subproblems as in predictive coding (PC) theory, which adds top-down connections between consecutive layers. In this study, we introduce a new model, 2-layer sparse predictive coding (2L-SPC), to assess the impact of this interlayer feedback connection. In particular, the 2L-SPC is compared with a hierarchical Lasso (Hi-La) network made out of a sequence of independent Lasso layers. The 2L-SPC and a 2-layer Hi-La networks are trained on four different databases and with different sparsity parameters on each layer. First, we show that the overall prediction error generated by 2L-SPC is lower thanks to the feedback mechanism as it transfers prediction error between layers. Second, we demonstrate that the inference stage of the 2L-SPC is faster to converge and generates a refined representation in the second layer compared to the Hi-La model. Third, we show that the 2L-SPC top-down connection accelerates the learning process of the HSC problem. Finally, the analysis of the emerging dictionaries shows that the 2L-SPC features are more generic and present a larger spatial extension.

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71. Resonator Networks, 1: An Efficient Solution for Factoring High-Dimensional, Distributed Representations of Data Structures

E. Paxon Frady, Spencer J. Kent, Bruno A. Olshausen, Friedrich T. Sommer

https://doi.org/10.1162/neco_a_01331 | pp. 2311-2331

The ability to encode and manipulate data structures with distributed neural representations could qualitatively enhance the capabilities of traditional neural networks by supporting rule-based symbolic reasoning, a central property of cognition. Here we show how this may be accomplished within the framework of Vector Symbolic Architectures (VSAs) (Plate, 1991; Gayler, 1998; Kanerva, 1996), whereby data structures are encoded by combining high-dimensional vectors with operations that together form an algebra on the space of distributed representations. In particular, we propose an efficient solution to a hard combinatorial search problem that arises when decoding elements of a VSA data structure: the factorization of products of multiple codevectors. Our proposed algorithm, called a resonator network, is a new type of recurrent neural network that interleaves VSA multiplication operations and pattern completion. We show in two examples-parsing of a tree-like data structure and parsing of a visual scene-how the factorization problem arises and how the resonator network can solve it. More broadly, resonator networks open the possibility of applying VSAs to myriad artificial intelligence problems in real-world domains. The companion article in this issue (Kent, Frady, Sommer, & Olshausen, 2020) presents a rigorous analysis and evaluation of the performance of resonator networks, showing it outperforms alternative approaches.

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72. Resonator Networks, 2: Factorization Performance and Capacity Compared to Optimization-Based Methods

Spencer J. Kent, E. Paxon Frady, Friedrich T. Sommer, Bruno A. Olshausen

https://doi.org/10.1162/neco_a_01329 | pp. 2332-2388

We develop theoretical foundations of resonator networks, a new type of recurrent neural network introduced in Frady, Kent, Olshausen, and Sommer (2020), a companion article in this issue, to solve a high-dimensional vector factorization problem arising in Vector Symbolic Architectures. Given a composite vector formed by the Hadamard product between a discrete set of high-dimensional vectors, a resonator network can efficiently decompose the composite into these factors. We compare the performance of resonator networks against optimization-based methods, including Alternating Least Squares and several

gradient-based algorithms, showing that resonator networks are superior in several important ways. This advantage is achieved by leveraging a combination of nonlinear dynamics and searching in superposition, by which estimates of the correct solution are formed from a weighted superposition of all possible solutions. While the alternative methods also search in superposition, the dynamics of resonator networks allow them to strike a more effective balance between exploring the solution space and exploiting local information to drive the network toward probable solutions. Resonator networks are not guaranteed to converge, but within a particular regime they almost always do. In exchange for relaxing the guarantee of global convergence, resonator networks are dramatically more effective at finding factorizations than all alternative approaches considered.

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73. Differential Covariance: A New Method to Estimate Functional Connectivity in fMRI

Tiger W. Lin, Yusi Chen, Qasim Bukhari, Giri P. Krishnan, Maxim Bazhenov, Terrence J. Sejnowski
https://doi.org/10.1162/neco_a_01323 | pp. 2389-2421

Measuring functional connectivity from fMRI recordings is important in understanding processing in cortical networks. However, because the brain's connection pattern is complex, currently used methods are prone to producing false functional connections. We introduce differential covariance analysis, a new method that uses derivatives of the signal for estimating functional connectivity. We generated neural activities from dynamical causal modeling and a neural network of Hodgkin-Huxley neurons and then converted them to hemodynamic signals using the forward balloon model. The simulated fMRI signals, together with the ground-truth connectivity pattern, were used to benchmark our method with other commonly used methods. Differential covariance achieved better results in complex network simulations. This new method opens an alternative way to estimate functional connectivity.

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74. Synchrony and Complexity in State-Related EEG Networks: An Application of Spectral Graph Theory

Amir Ghaderi, Bianca R. Baltaretu, Masood Nemati Andevari, Vishal Bharmauria, Fuat Balc■
https://doi.org/10.1162/neco_a_01327 | pp. 2422-2454

The brain may be considered as a synchronized dynamic network with several coherent dynamical units. However, concerns remain whether synchronizability is a stable state in the brain networks. If so, which index can best reveal the synchronizability in brain networks? To answer these questions, we tested the application of the spectral graph theory and the Shannon entropy as alternative approaches in neuroimaging. We specifically tested the alpha rhythm in the resting-state eye closed (rsEC) and the resting-state eye open (rsEO) conditions, a well-studied classical example of synchrony in neuroimaging EEG. Since the synchronizability of alpha rhythm is more stable during the rsEC than the rsEO, we hypothesized that our suggested spectral graph theory indices (as reliable measures to interpret the synchronizability of brain signals) should exhibit higher values in the rsEC than the rsEO condition. We performed two separate analyses of two different datasets (as elementary and confirmatory studies). Based on the results of both studies and in agreement with our hypothesis, the spectral graph indices revealed higher stability of synchronizability in the rsEC condition. The k-mean analysis indicated that the spectral graph indices can distinguish the rsEC and rsEO conditions by considering the synchronizability of brain

networks. We also computed correlations among the spectral indices, the Shannon entropy, and the topological indices of brain networks, as well as random networks. Correlation analysis indicated that although the spectral and the topological properties of random networks are completely independent, these features are significantly correlated with each other in brain networks. Furthermore, we found that complexity in the investigated brain networks is inversely related to the stability of synchronizability. In conclusion, we revealed that the spectral graph theory approach can be reliably applied to study the stability of synchronizability of state-related brain networks.

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75. Toward a Unified Framework for Cognitive Maps

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https://doi.org/10.1162/neco_a_01326 | pp. 2455-2485

In this study, we integrated neural encoding and decoding into a unified framework for spatial information processing in the brain. Specifically, the neural representations of self-location in the hippocampus (HPC) and entorhinal cortex (EC) play crucial roles in spatial navigation. Intriguingly, these neural representations in these neighboring brain areas show stark differences. Whereas the place cells in the HPC fire as a unimodal function of spatial location, the grid cells in the EC show periodic tuning curves with different periods for different subpopulations (called modules). By combining an encoding model for this modular neural representation and a realistic decoding model based on belief propagation, we investigated the manner in which self-location is encoded by neurons in the EC and then decoded by downstream neurons in the HPC. Through the results of numerical simulations, we first show the positive synergy effects of the modular structure in the EC. The modular structure introduces more coupling between heterogeneous modules with different periodicities, which provides increased error-correcting capabilities. This is also demonstrated through a comparison of the beliefs produced for decoding two- and four-module codes. Whereas the former resulted in a complete decoding failure, the latter correctly recovered the self-location even from the same inputs. Further analysis of belief propagation during decoding revealed complex dynamics in information updates due to interactions among multiple modules having diverse scales. Therefore, the proposed unified framework allows one to investigate the overall flow of spatial information, closing the loop of encoding and decoding self-location in the brain.

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76. Active Learning for Level Set Estimation Under Input Uncertainty and Its Extensions

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https://doi.org/10.1162/neco_a_01332 | pp. 2486-2531

Testing under what conditions a product satisfies the desired properties is a fundamental problem in manufacturing industry. If the condition and the property are respectively regarded as the input and the output of a black-box function, this task can be interpreted as the problem called level set estimation (LSE): the problem of identifying input regions such that the function value is above (or below) a threshold. Although various methods for LSE problems have been developed, many issues remain to be solved for their practical use. As one of such issues, we consider the case where the input conditions cannot be controlled precisely-LSE problems under input uncertainty. We introduce a basic framework for handling input uncertainty in LSE problems and then propose efficient methods with proper theoretical guarantees.

The proposed methods and theories can be generally applied to a variety of challenges related to LSE under input uncertainty such as cost-dependent input uncertainties and unknown input uncertainties. We apply the proposed methods to artificial and real data to demonstrate their applicability and effectiveness.

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77. Redundancy-Aware Pruning of Convolutional Neural Networks

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https://doi.org/10.1162/neco_a_01330 | pp. 2532-2556

Pruning is an effective way to slim and speed up convolutional neural networks. Generally previous work directly pruned neural networks in the original feature space without considering the correlation of neurons. We argue that such a way of pruning still keeps some redundancy in the pruned networks. In this letter, we proposed to prune in the intermediate space in which the correlation of neurons is eliminated. To achieve this goal, the input and output of a convolutional layer are first mapped to an intermediate space by orthogonal transformation. Then neurons are evaluated and pruned in the intermediate space. Extensive experiments have shown that our redundancy-aware pruning method surpasses state-of-the-art pruning methods on both efficiency and accuracy. Notably, using our redundancy-aware pruning method, ResNet models with three times the speed-up could achieve competitive performance with fewer floating point operations per second even compared to DenseNet.

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78. Analyzing and Accelerating the Bottlenecks of Training Deep SNNs With Backpropagation

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https://doi.org/10.1162/neco_a_01319 | pp. 2557-2600

Spiking neural networks (SNNs) with the event-driven manner of transmitting spikes consume ultra-low power on neuromorphic chips. However, training deep SNNs is still challenging compared to convolutional neural networks (CNNs). The SNN training algorithms have not achieved the same performance as CNNs. In this letter, we aim to understand the intrinsic limitations of SNN training to design better algorithms. First, the pros and cons of typical SNN training algorithms are analyzed. Then it is found that the spatiotemporal backpropagation algorithm (STBP) has potential in training deep SNNs due to its simplicity and fast convergence. Later, the main bottlenecks of the STBP algorithm are analyzed, and three conditions for training deep SNNs with the STBP algorithm are derived. By analyzing the connection between CNNs and SNNs, we propose a weight initialization algorithm to satisfy the three conditions. Moreover, we propose an error minimization method and a modified loss function to further improve the training performance. Experimental results show that the proposed method achieves 91.53% accuracy on the CIFAR10 data set with 1% accuracy increase over the STBP algorithm and decreases the training epochs on the MNIST data set to 15 epochs (over 13 times speed-up compared to the STBP algorithm). The proposed method also decreases classification latency by over 25 times compared to the CNN-SNN conversion algorithms. In addition, the proposed method works robustly for very deep SNNs, while the STBP algorithm fails in a 19-layer SNN.